Anikait Singh

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Research Interests

My research develops efficient adaptation algorithms for open-ended discovery, with the long-term goal of building AI systems that can make progress on problems such as scientific research. Large-scale pre-training and post-training have given foundation models broad world knowledge and strong reasoning priors, yet adapting to novel environments remains a bottleneck when tasks require new knowledge or behaviors. I study how AI systems can adapt efficiently through interaction, developing algorithms for three complementary capabilities: exploring new behaviors, incorporating new knowledge, and uncovering the underlying structure of an environment. Together, these capabilities can form a flywheel for improvement in open-ended settings, where agents synthesize experience and new knowledge gathered through environmental interaction to direct future exploration.

Education

Stanford University, Palo Alto, CA — Sep 2023 – Present
Ph.D. in Computer Science — Advisors: Prof. Chelsea Finn and Aviral Kumar

University of California, Berkeley, Berkeley, CA — Aug 2019 – May 2023
B.A. in Computer Science (Highest Distinction) — GPA: 3.986; Technical GPA: 4.0
Selected coursework: Deep Learning, Reinforcement Learning, Convex Optimization, NLP, and Info Theory.

Experience

Meta Superintelligence Labs, Paris, France — Jun 2026 – October 2026
Research Scientist Intern.

Microsoft Research, New York, NY — Jun 2025 – Sep 2025
Research Scientist Intern.

Synthlabs AI, San Francisco, CA — Jan 2025 – March 2025
Research Collaboration.

Toyota Research Institute (TRI), Los Altos, CA — Jun 2024 – Sep 2024
Research Scientist Intern.

Google DeepMind Robotics, Mountain View, CA — Apr 2023 – Feb 2024
Student Researcher.

X, the moonshot factory, Mountain View, CA — Dec 2022 – Apr 2023
AI Resident.

Selected Publications

(* equal contribution; selection)

- **Learning to Discover Abstractions for LLM Reasoning**. Y. Qu*, **A. Singh***, Y. Lee*, A. Setlur, R. Salakhutdinov, C. Finn, A. Kumar. ICLR 2026. ICML 2025 ES-FoMo Workshop (Spotlight).
- **Meta Rubrics: Learning to Discover Rubrics at Deployment for Efficient Task Adaptation**. **A. Singh***, A. Kumar, C. Finn, R. Munos. Under Submission to ICLR 2027.
- **Librarian Agents: Training LLMs to Curate Workspaces that Anticipate Future Queries**. **A. Singh***, T. Zhang*, Y. Lee*, A. Kumar, C. Finn. Under Submission to ICLR 2027.
- **GIANTS: Generative Insight Anticipation from Scientific Literature**. J. He-Yueya*, **A. Singh***, G. Gao, M. Y. Li, S. Yang, C. Finn, E. Brunskill, N. D. Goodman. CoLM 2026 LM4Sci Workshop (Outstanding Paper Award). Under Submission to ICLR 2027.

- **Few-Shot Preference Optimization of Synthetic Preferences Elicits Effective Personalization to Real Users**. **A. Singh***, S. Hsu*, E. Mitchell, T. Hashimoto, A. Sharma, C. Finn. ICLR 2026.
- **Cognitive Behaviors that Enable Self-Improving Reasoners**. K. Gandhi, A. Chakravarthy, **A. Singh**, N. Lile, N. D. Goodman. CoLM 2025 (Oral).
- **Improving Test-Time Search in LLMs with Backtracking Against In-Context Value Verifiers**. **A. Singh**, K. Arora, S. Keh, T. Hashimoto, C. Finn, A. Kumar. ICLR 2025 Reasoning and Planning for LLMs Workshop (Best Paper Runner-Up).
- **Personalized Preference Fine-tuning of Diffusion Models**. M. Dang*, **A. Singh***, L. Zhou, S. Ermon, J. Song. CVPR 2025.
- **Towards System 2 Reasoning in LLMs: Learning How to Think With Meta Chain-of-Thought**. V. Xiang, C. Snell, K. Gandhi, A. Albalak, **A. Singh**, C. Blagden, D. Phung, R. Rafailov, N. Lile, D. Mahan, L. Castricato, J.-P. Franken, N. Haber, C. Finn. Position Paper, 2024.
- **Preference Fine-Tuning of LLMs Should Leverage Suboptimal, On-Policy Data**. **A. Singh***, F. Tajwar*, A. Sharma, R. Rafailov, J. Schneider, T. Xie, S. Ermon, C. Finn, A. Kumar. ICML 2024.
- **Robotic Offline RL from Internet Videos via Value-Function Pre-Training**. C. Bhateja*, D. Guo*, D. Ghosh*, **A. Singh**, M. Tomar, Q. Vuong, Y. Chebotar, S. Levine, A. Kumar. ICRA 2024.
- **Offline RL With Realistic Datasets: Heteroskedasticity and Support Constraints**. **A. Singh***, A. Kumar*, Q. Vuong, Y. Chebotar, S. Levine. NeurIPS 2023.
- **Cal-QL: Calibrated Offline RL Pre-Training for Efficient Online Fine-Tuning**. M. Nakamoto*, Y. Zhai*, **A. Singh**, M. S. Mark, Y. Ma, C. Finn, A. Kumar, S. Levine. NeurIPS 2023.
- **Pre-Training for Robots: Offline RL Enables Learning from a Handful of Trials**. A. Kumar*, **A. Singh***, F. Ebert*, M. Nakamoto, Y. Yang, C. Finn, S. Levine. RSS 2023.
- **RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control**. Google DeepMind Robotics. ICRA 2023.
- **When Should We Prefer Offline Reinforcement Learning Over Behavioral Cloning?**. A. Kumar, J. Hong, **A. Singh**, S. Levine. ICLR 2022.
- **A Workflow for Offline Model-Free Robotic Reinforcement Learning**. A. Kumar*, **A. Singh***, S. Tian, C. Finn, S. Levine. CoRL 2021 (Oral).

Teaching

Mentoring Program Organizer, Stanford CS Undergraduate Mentoring Program (2024-Present).
Organizer, Stanford NLP Seminar (2025-2026).
Program Coordinator & Mentor, Deep Learning Portal (2024).
Graduate Student Instructor, CS224r (Spring 2025/2026, Project Lead).

Honors

NSF Graduate Research Fellowship (2023–2028).
Gemini Academic Program Award (2025)
Tinker & Modal Academic Research Grants (2025/2026)

Invited Talks and Service

Invited Talks at Engram, Salesforce Future AI Forum, Jump Trading, and ServiceNow, BeNeRL
Reviewer (w/honors) for ICLR, NeurIPS, ICML, CoRL, ICRA (RAL), AAAI, and CoLM.